Woods & manufactured boards

WOODWORK WITH MR KELLY

National 5 · Machines, materials & finishing · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ identify and choose softwoods: pine, cedar, larch
- ☐ identify and choose hardwoods: ash, oak, beech, mahogany, meranti
- ☐ identify manufactured boards from their edges and give their uses
- ☐ justify a material choice by linking a property to the use

I DO — Worked example

Read how a full-mark answer is built

A garden chair for children will be left outdoors.

1 Identify an appropriate wood for the chair and a property that makes it suitable. (2 marks)

Full-mark answer each ✓ is one mark

Cedar ✓. Its natural oils make it durable outdoors, so it resists rot in the rain ✓.

Why marks are not awarded

"Pine because it is cheap."

0 marks: pine isn't durable outdoors without treatment, and "cheap" isn't linked to outdoor use.

WE DO — Guided practice

Use the starters and word bank to help you

A mallet head needs to be hard and resist splitting.

2 Name a suitable hardwood. (1 mark)

Sentence starters

1. …

Word bank

Tip: 1 marks = 1 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which board has layers with the grain at 90° to each other?

☐ MDF ☐ Chipboard ☐ Plywood ☐ Hardboard

b) Which is an advantage of manufactured boards?

☐ Attractive grain ☐ Available in large, stable sheets ☐ Always waterproof ☐ Always stronger than solid wood

c) Which board's edges crumble and are best joined with KD fittings?

☐ Plywood ☐ Blockboard ☐ Chipboard ☐ Oak

d) Which is a hardwood?
☐ Larch ☐ Pine ☐ Beech ☐ Cedar

4 Key terms: write the letter of the matching description next to each term.

1. **Softwood**	____	**A** Tough and flexible: tool handles
2. **Hardwood**	____	**B** From broad-leaved trees: grows slowly, usually harder and dearer
3. **Cedar property**	____	**C** Hard and close grained: turns well, mallets
4. **Ash property**	____	**D** Natural oils make it durable outdoors
5. **Beech property**	____	**E** From conifers: grows fast, cheaper, more sustainable
6. **Plywood**	____	**F** Veneers with grain at 90°: strong in all directions

A pupil is shown the edges of three manufactured boards.

5 Describe how you would identify plywood from its edge. (1 mark)

A drawer base is made from a manufactured board.

6 Name a suitable board and give a reason for your choice. (2 marks)

A table top could be made from oak or pine. Their data is given in a table.

7 Explain which wood is better for a hard-wearing table top. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?